PROJECT REPORT: "AI-Enhanced Wildlife Corridor Identification using A* Search"

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Course: CSE-UCS-1002: Introduction to AI-ML

1. Introduction (Project Knowledge)

Human growth, like the development of cities, highways, and farms, tends to interfere with habitats. The process splits large, contiguous wildlife habitats into smaller, fragmented portions. Fragmentation threatens biodiversity by restricting animal movement, impeding access to resources, and reducing genetic diversity.

Wildlife corridors are narrow strips of natural landscape joining these fragmented spaces. They are important for conservation, as they enable animals to travel freely between areas in quest of food, mates, and new habitats. Choosing the optimal corridor route is difficult, though. The optimal route is not necessarily the shortest route; it should be safe and accessible for the species in question.

This project attempts to address this challenge by using the A* (A-Star) search algorithm, an important Artificial Intelligence concept, to determine the optimal, minimum-cost path for a wildlife corridor over a simulated environment.

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2. Methodology (Application of AI Concepts & Algorithm)

In order to find the optimal corridor, we created the landscape and implemented the A* algorithm.

A. Environment Representation

It was modeled as a 2D grid, with each cell mapping to a particular area of land. We gave each cell a traversal cost in order to assess how well the land is suited to wildlife:

- Low Cost (e.g., 1): Good terrain, such as forests or grass.
- Medium Cost (e.g., 10): Less ideal terrain, such as hills or thick scrub.
- High Cost (i.e., 100): Hazardous or inaccessible regions, such as city areas, significant roads, or significant bodies of water.

NOTE : The aim of the algorithm is to locate a smooth path from an initiation point (Habitat A) to a destination point (Habitat B) with minimum total costs.

B. The A* Search Algorithm

A* is a search algorithm for pathfinding with a guidance. It finds the minimum-cost path efficiently by employing a heuristic to guide its search. The heart of the algorithm consists

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of its evaluation function for any node (grid cell) n:

Where:

- g(n) is the true cost from the initial point to node n.
- h(n) is an estimate of the cost of the shortest path from node n to the goal. For this map with a grid-based layout, Manhattan Distance was used as the heuristic. It is admissible since it never overestimates the true cost, so A* will find the optimal solution.
- f(n) is the sum of the estimated cost of the path via node n.

A* has an "Open List" of nodes to visit and a "Closed List" of nodes that have been visited. It keeps picking the node with the lowest score in the Open List until it finds the destination.

3. Implementation and Results (Coding & Visualization)

It was implemented using Python 3 with the following libraries:

- NumPy: To effectively create and handle the 2D landscape grid.
- Matplotlib: To graphically visualize the landscape, obstacles, and the resultant corridor computed.

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NOTE: The software initially makes a cost map to map the simulated landscape.

Subsequently, it applies the A* algorithm to compute the shortest cost path between the specified start and end points.

The output produces a graphical view wherein:

1. The terrain is color-coded according to travel cost.

2. The origin and destination are labeled clearly.

3. The most cost-effective path, which has been labeled as the wildlife corridor, is represented as a clear line.

This route avoids successfully danger zones of high cost, instead opting to cut through lower-cost safer land, even if it means a slightly longer path. This demonstrates the flexibility of the algorithm in balancing safety and distance to build a viable corridor.

4. Conclusion

This project successfully illustrated the application of the A* search algorithm to optimize wildlife corridor location in a simulation scenario. By reducing an intricate ecological issue to a pathfinding exercise in a cost-grid, we can employ AI to deliver well-informed and practical recommendations for conservation planning. This building block could be built upon by using actual GIS data, implementing additional sophisticated cost

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factors, and combining machine learning models to forecast animal movement behavior.

5. References (Gathering Resources)

1. Russell, S. J., & Norvig, P. (2020). Artificial Intelligence: A Modern Approach. Pearson.

2. Hart, P. E., Nilsson, N. J., & Raphael, B. (1968). A Formal Basis for the Heuristic Determination of Minimum Cost Paths. IEEE Transactions on Systems Science and Cybernetics, 4(2), 100-107.

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